Set C

a)In a disaster management scenario, researchers need to compare the DNA sequences of two different viral strains to find the longest shared genetic pattern (subsequence). Unlike a substring, a subsequence does not need to be contiguous but must appear in the same relative order.

Write a program using **Dynamic Programming** to find the length of the Longest Common Subsequence between two strings.

Input Format:

- First line: String s1
- Second line: String s2

Test Cases

Input:
abcbdab
bdcaba
Output:
bcba
bcab
bdab

Input:
XMJYAUZ
MZJAWXU
Output:
MJAU
MZAU

b) A highly relevant modification for the **longest common subsequence (lcs)** problem, specifically within the context of **drone-based disaster management**, is the **weighted lcs**. In a standard **lcs**, every matching character is treated with equal importance. However, in disaster management, certain coordinate signals may have higher priority (weight) than others. You are given two sequences S1 and S2 representing drone sensor signal streams, and a weight array W where each character has an associated priority weight.

Modify the same DP algorithm in the (a) part to compute the **maximum total weight** of the longest common subsequence between S1 and S2. If multiple sequences are there of same length only print the subsequence giving highest weight.

Test cases
Input:
rescue
50 10 20 30 15 10
secure
20 10 30 15 50 10
Output:
max weight: 85
subsequence: ecue

Input:

ABCDEF
2 3 4 5 6 7
CFDEPQ
4 7 5 6 9 1
Output:
max weight: 15
subsequence: CDE

Input
abcbdab
1 2 3 2 4 1 2
bdcaba
2 4 3 1 2 1
output
max weight: 9
subsequence: bdab